Sequences and Series: Arithmetic Sequences

Questions 1–52

Question 1. Write the nth term, u_n, for each of these arithmetic sequences in the form

$$u_n = a + (n-1)d$$

where a is the first term and d is the common difference.

a) $3, 8, 13, 18, 23, \ldots$

b) $25, 35, 45, 55, 65, \ldots$

c) $12, 6, 0, -6, -12, \ldots$

d) $150, 135, 120, 105, 90, \ldots$

e) $-44, -41, -38, -35, -32, \ldots$

f) $-100, -121, -142, -163, -184, \ldots$

g) $\frac{5}{6}, \frac{31}{30}, \frac{37}{30}, \frac{43}{30}, \frac{49}{30}, \ldots$

h) $\frac{11}{4}, -\frac{7}{4}, -\frac{25}{4}, -\frac{43}{4}, -\frac{61}{4}, \ldots$

Question 2. It is given that $9, 12, 15, 18, 21, \ldots$ is an arithmetic sequence. Find the 500th term in the sequence.

Question 3. It is given that $\frac{3}{2}, \frac{11}{6}, \frac{13}{6}, \frac{5}{2}, \frac{17}{6}, \ldots$ is an arithmetic sequence. Find the 240th term in the sequence.

Question 4. It is given that $-50, -45, -40, -35, \ldots, 765$ is a finite arithmetic sequence. Determine how many terms there are in the sequence.

Question 5. It is given that $31.3, 31.6, 31.9, 32.2, \ldots, 117.7$ is a finite arithmetic sequence. Determine how many terms there are in the sequence.

Question 6. It is given that $58.2, 57.5, 56.8, 56.1, \ldots, -25.1$ is a finite arithmetic sequence. Determine how many terms there are in the sequence.

Question 7. An arithmetic progression has 3rd term 22 and 7th term 54. Find the first term and the common difference.

Question 8. An arithmetic progression has 5th term 38 and 10th term 50.5. Find the first term and the common difference.

Question 9. An arithmetic progression has 2nd term 71 and 9th term 631. Find the first term and the common difference.

Question 10. An arithmetic progression has 6th term 20 and 12th term 38. Find the first term that exceeds 100.

Question 11. An arithmetic sequence has 3rd term $2 + \sqrt{3}$ and 9th term $3 + \sqrt{5}$. Find the common difference in the form $a + b\sqrt{3} + c\sqrt{5}$, where a, b and c are rational numbers.

Question 12. A botanist is studying the growth of a particular species of plant. The height of the plant after 2 weeks is 15 cm, and after 7 weeks it is 55 cm. Given the height of the plant over the weeks follows an arithmetic sequence, find the height of the plant after 3 weeks.

Question 13. An arithmetic sequence has the three consecutive terms $\ln \frac{1}{4}k$, $\ln \frac{1}{3}k$, $\ln p$, where k, p are positive constants. Find p in terms of k.

Question 14. An arithmetic sequence has the three consecutive terms $6k^2$, $5k$, $(21k - 35)$, where k is a constant. Find the possible values of k.

Question 15. An arithmetic sequence has the three consecutive terms $\frac{2}{k}$, $20k$, $-(5k+13)$, where k is a constant. Find the possible values of k.

Question 16. An arithmetic sequence has the three consecutive terms $16e^{k}$, 20, $16e^{-k}$, where k is a constant. Find the possible values of k.

Question 17. An arithmetic sequence has the three consecutive terms 28, $\frac{200}{2^k}$, $\left(\frac{240}{2^k} - 2^k\right)$, where k is a constant. Find the possible values of k.

Question 18. The first three terms of an arithmetic sequence are $32 + 3k$, $13 - 4k$, $78 + 10k$, where k is a constant.

a) Find the value of k.

b) Find the 100th term in the sequence.

Question 19. Find the sum of the arithmetic series $3 + 11 + 19 + 27 + \cdots + 155$.

Question 20. Find the sum of the arithmetic series $14 + 20 + 26 + 32 + \cdots + 1238$.

Question 21. Find the sum of the arithmetic series $-11 - 20 - 29 - 38 - \cdots - 542$.

Question 22. Find the sum of the arithmetic series $\frac{20}{3} + \frac{35}{6} + 5 + \frac{25}{6} + \cdots - 60$.

Question 23. An arithmetic series is given by $\sum_{i=7}^{40}\left(6i-\frac{1}{2}\right)$.

a) Write down the first term and common difference of the series.

b) Write down how many terms there are in the series.

Question 24. An arithmetic sequence has rth term given by $u_r = 7 - 10r$. Show that $\sum_{r=1}^{40} u_r = -7920$.

Question 25. An arithmetic sequence has kth term given by $u_k = -3 + \frac{k}{3}$. Show that $\sum_{k=10}^{30} u_k = 77$.

Question 26. Find the sum of the first 150 terms of the arithmetic sequence $25 + 26.5 + 28 + 29.5 + \cdots$.

Question 27. Find the sum of the first 80 terms of the arithmetic sequence $\frac{1}{5} + \frac{7}{5} + \frac{13}{5} + \frac{19}{5} + \cdots$.

Question 28. Find the sum of the first 185 terms of the arithmetic sequence $100 + 99.2 + 98.4 + 97.6 + \cdots$.

Question 29. Find the sum of the first 390 terms of the arithmetic sequence $10.3 + 10 + 9.7 + 9.4 + \cdots$.

Question 30. On his 1st birthday, Jaime's parents put £35 into a bank account. On his 2nd birthday, Jaime's parents put £65 into the bank account. The amounts deposited into the bank account form an arithmetic sequence. Determine the total amount of money in the bank account the day after Jaime's 18th birthday.

Question 31. It is given that S_n is the sum of the first n terms of the arithmetic sequence $104 + 101 + 98 + 95 + \cdots$. Find the maximum value of S_n.

Question 32. An arithmetic sequence has nth term given by $u_n = \frac{6n}{5} + \frac{2}{3}$. Find the value of k such that

$$\sum_{n=1}^{k} u_n < 200 \quad \text{and} \quad \sum_{n=1}^{k+1} u_n > 200.$$

Question 33. An arithmetic sequence has nth term given by $u_n = 4.1n - 3$. Find the value of k such that

$$\sum_{n=1}^{k} u_n < 3000 \quad \text{and} \quad \sum_{n=1}^{k+1} u_n > 3000.$$

Question 34. Aiden buys a new phone on a contract that is worth £900 at 0% interest. The contract provider requires Aiden to pay back £10.50 at the end of week 1, £10.60 at the end of week 2, £10.70 at the end of week 3, etc, until the £900 is fully paid off, forming an arithmetic sequence. At the end of the final week, Aiden pays off the remaining balance if it is less than the amount he paid the previous week.

a) Determine how many weeks it will take to pay off the contract.

b) Determine how much Aiden pays on the final week.

Question 35. An arithmetic series has 3rd term 46 and 15th term 222. Find the sum of the first 30 terms.

Question 36. An arithmetic series has 1st term 3. The sum of the first 100 terms is 12. Find the common difference.

Question 37. The first three terms of an arithmetic progression are $8k$, $k^2(k-4)$, 10, where k is a real constant. It is given that the sum of the first n terms is -53900. Find the value of n.

Question 38. An arithmetic sequence has first term a and common difference d. The sum of the first 15 terms is -465.

a) Show that $a + 7d = -31$.

It is given that the 50th term is -283.

b) Find the value of a and the value of d.

Question 39. An arithmetic sequence has first term a and common difference d. The sum of the first 10 terms is 410.

a) Show that $2a + 9d = 82$.

It is given that the sum of the first 20 terms is 1620.

b) Find the value of a and the value of d.

c) Find the sum of the first 30 terms.

Question 40. An arithmetic sequence has first term a and common difference d. The sum of the first 490 terms is 1489 times the sum of the first 10 terms.

a) Show that $3a = 11d$.

b) Given that the 10th term is 38, find the value of a and the value of d.

Question 41. An arithmetic progression has first term a and common difference d. The sum of the first 220 terms is 211 times the sum of the first 20 terms.

a) Show that $a + 4d = 0$.

b) Given that the 105th term is -200, find the value of a and the value of d.

Question 42. An arithmetic sequence has first term a and common difference d. The sum of the first 425 terms is 221 times the square of the sum of the first 5 terms.

a) Show that $221a^2 - 17a + 884d^2 - 3604d + 884ad = 0$.

b) Given that the 2nd term is 4 and that both a and d are integers, find the value of a and the value of d.

Question 43. The first three terms of an arithmetic sequence are given by $\frac{1}{k-4}$, $-\frac{1}{k}$ and $\frac{1}{k+4}$, where k is a positive constant. Show that the sum of the first 52 terms is $a + b\sqrt{2}$, where a and b are integers.

Question 44. The first three terms of an arithmetic sequence are 8, 11, and 14. The sum of the first n terms is greater than 1000 but less than 2000. Find the possible values of n.

Question 45. An arithmetic series with sum 100 is given by $\sum_{n=1}^{100}(pn+q)$ where p and q are constants. It is also given that $u_{51}+u_{53}+u_{55}=u_{59}$. Find the value of p and the value of q.

Question 46. An arithmetic sequence has first term $(3k+2)$ and last term $(9k-1)$, where k is a positive constant. The common difference of the sequence is 6. Find the number of terms in the sequence in terms of k.

Question 47. It is given that k is a positive integer. A_k is the sum of the first k terms of an arithmetic series with first term 3 and common difference 8. B_k is the sum of the first k terms of an arithmetic series with first term -52 and common difference 18. Find the value of k such that $A_k = B_k$.

Question 48. It is given that k is a positive integer. C_k is the sum of the first k terms of an arithmetic series with first term 1500 and common difference -9. D_k is the sum of the first k terms of an arithmetic series with first term 6 and common difference 3. Find the value of k such that $C_k = D_k$.

Question 49. An arithmetic sequence has first term a and common difference d. Show that S_n, the sum of the first n terms, is $\frac{1}{2}n(2a + (n-1)d)$.

Question 50. Find the sum of all the even numbers from 100 to 500.

Question 51. Find the sum of all the odd numbers from 5 to 555.

Question 52. It is given that $u_1 = p$, $u_2 = (2p-1)$, $u_3 = (3p+2)$, $u_4 = (15p+\frac{2}{3})$, ... is **not** an arithmetic sequence, where p is a constant such that $p \geq 1$. However, the sequence formed by the ratios $\frac{u_{n+1}}{u_n}$ is an arithmetic sequence. Find the value of p and hence find the value of u_5.